

When Certainty Is Not Worth It: Capital Lock-Up and Settlement Discounting in Prediction Markets

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Abstract

Collateralized prediction markets are contingent-claim markets in which economic uncertainty can disappear before winning claims become redeemable. This paper studies the pricing effect of that delay. When collateral remains locked until oracle settlement, a near-certain dollar is a delayed dollar, so prices embed a maturity-dependent settlement discount in addition to beliefs about outcomes. We recover an implied settlement-discount term structure from persistent near-certain contracts using realized settlement times and summarize it as an annualized settlement wedge (ASW). The recovered wedges are positive, maturity-dependent, and time-varying. Adjusting prices by these curves reduces the near-certainty horizon gradient by roughly 48–88%, indicating that much of the raw maturity pattern reflects priced settlement frictions rather than forecast error alone. Market architecture changes the wedge: negRisk conversion compresses discounts by recycling part of the position into synthetic collateral, while yield-bearing collateral flattens the term structure by reducing the opportunity cost of lock-up. The results show that pricing quality in prediction markets is endogenous to settlement mechanics, collateral productivity, and capital-recycling design. Prediction-market prices therefore aggregate information through a financial infrastructure whose funding conditions are measurable and economically important.

1 Introduction

Prediction-market prices are increasingly used as public probability signals. Odds from platforms such as Polymarket and Kalshi now appear in political commentary, media dashboards, and creator content, often as complements or substitutes for polls [HK26]. The standard logic is simple: a winner-take-all contract pays \$1 if an event occurs and \$0 otherwise, so its price can be read as the market-implied probability of the event [WZ04]. Much of the empirical literature adopts this mapping and interprets systematic deviations between prices and realized frequencies as informational or behavioral error, including miscalibration and favorite-longshot bias [Man06, PC13].

This paper argues that the mapping is incomplete for collateralized, oracle-settled prediction markets. The standard price-as-probability benchmark remains useful, but these venues are information-aggregation mechanisms embedded in capital markets rather than frictionless probability oracles. Their prices are discounted probabilities: prices of delayed contingent claims that combine beliefs with settlement delay, locked-capital opportunity cost, liquidity demand, and residual platform or oracle risk. Traders allocate collateral to state-contingent claims, and that collateral can remain locked even after the real-world uncertainty has effectively disappeared. Systematic price–frequency gaps may therefore reflect financial architecture as well as forecasting error.

The intuition is that a nearly certain dollar paid after delayed settlement is not the same object as an immediately redeemable dollar. Polymarket’s “Will Jesus Christ return in 2025?” market¹ is illustrative [Pol26c]: for months, the complementary near-certain NO side traded around \$0.96. A trader buying \$1,000 of NO at that price would earn roughly 4.2% if the position paid \$1 at settlement, but only after locking capital for most of the year. A below-par price can therefore be consistent with near certainty once outside returns, liquidity needs, and residual platform risk are priced in.

¹Polymarket market page: <https://polymarket.com/event/will-jesus-christ-return-in-2025>

We formalize this friction as settlement-induced discounting. Let $X_i \in \{0, 1\}$ denote the settlement payoff of contract i and let $\tau_{i,t}$ denote the remaining time until formal resolution. Instead of the frictionless benchmark $P_{i,t} = \mathbb{E}_t[X_i]$, we study the reduced-form pricing equation

$$P_{i,t} = \mathbb{E}_t[X_i]D(\tau_{i,t}),$$

where $D(\tau) \leq 1$ summarizes the market value of delayed settlement, capital lock-up, outside opportunities, liquidity, and residual platform risk. Observed prices are therefore discounted probabilities rather than frictionless probabilities; interpreting them as probabilities requires accounting for settlement discounting. The wedge should be largest at longer settlement horizons, should generate apparent underconfidence in near-certain contracts, and should shrink when market design allows capital to earn yield or be recycled earlier.

The empirical challenge is that high prices need not represent certainty: they may still contain residual belief uncertainty. We therefore use persistent upper-tail contracts as candidate support for recovering the discount component: markets in which one side remains highly priced for multiple consecutive days and does not subsequently reverse. We estimate a high-quantile near-certainty frontier by realized time to settlement and convert that frontier into an annualized settlement wedge (ASW), a reduced-form required-return measure for locked prediction-market capital.

We find a persistently positive and maturity-dependent ASW term structure. Discount adjustment reduces the near-certainty horizon gradient by roughly 48–88%, and event-level cross-fitting yields similar attenuation. Under the highest-wedge robust frontier considered, the residual gradient becomes statistically indistinguishable from zero. These results indicate that a substantial share of apparent long-horizon miscalibration reflects mechanical settlement discounting rather than purely informational error.

Relative to prior conceptual work on cost-of-capital channels [GJK19] and earlier evidence on horizon effects [PC13], our contribution is threefold:

1. We model collateralized prediction-market prices as discounted expected payoffs under delayed settlement and locked collateral.
2. We recover a frontier-implied ASW term structure using near-certain contracts and realized settlement times, and show that discount adjustment attenuates horizon-conditioned calibration distortion.
3. We show that market design changes the discount wedge: negRisk conversion mechanically compresses near-certainty discounts, and cross-platform evidence is consistent with yield-bearing collateral flattening the wedge.

Thus, forecast evaluation, cross-platform comparison, and market-design assessment must account for the financing environment in which prediction-market prices are formed.

2 Background

This section isolates the primitives needed for the pricing mechanism. The causal chain is simple: economically resolved claims can remain formally unsettled; unsettled claims keep capital from being redeployed; and market architecture determines whether that lock-up is mitigated through conversion or collateral yield.

2.1 Settlement Timing and Locked Collateral

The wedge starts with a distinction between knowing an outcome and redeeming a claim. For each market i , let T_i^{event} denote the time at which the outcome is effectively determined from publicly available information; T_i^{end} the platform’s scheduled market end or trading-close metadata; T_i^{oracle} the time at which the oracle outcome becomes finalized; and T_i^{settle} the first time at which winning claims can be redeemed. These timestamps need not coincide: T_i^{event} can precede T_i^{end} , and both can precede oracle finalization and redemption.

On Polymarket, resolution is handled through the UMA Optimistic Oracle, under which anyone can propose an outcome, undisputed proposals finalize after a two-hour challenge window, and disputed

cases can escalate to UMA’s voting mechanism [Pol26d, UMA26]. This architecture creates an oracle stage between economic resolution and redeemability: even after the outcome is clear, claims become cash only when the oracle path finalizes. Empirically, we proxy settlement completion using realized oracle-finalization timestamps, since redemption becomes possible only after oracle resolution.

In the framework below, $T_i^{\text{resolution}}$ refers to T_i^{settle} , not the scheduled market end. The economic implication is direct: capital tied to unresolved claims cannot be redeployed until redemption and therefore forgoes outside returns. Claimholders may also bear liquidity costs and residual oracle or platform risk during the settlement interval. A near-certain \$1 payoff is therefore a delayed claim rather than an immediately spendable dollar.

2.2 Contract Architecture

A binary prediction contract is a state-contingent claim paying \$1 if an event occurs and \$0 otherwise. On decentralized venues, such claims are often implemented through the Gnosis Conditional Token Framework [Gno20] (CTF): collateral is locked, a complete set of outcome-contingent claims is minted, and that set can be split into tradable outcome tokens. A complete set sums to par at resolution, while losing outcome tokens expire worthless.

Only one feature of this architecture is essential for the model: unless the position can be merged, converted, or otherwise recycled, the collateral value embedded in unresolved outcome tokens remains unavailable until settlement. Protocol operations that recombine claims into complete-set or collateral-equivalent positions therefore matter because they shorten the effective lock-up even when the formal settlement date is unchanged.

2.3 Capital-Efficient Market Design

Market design changes the wedge by changing either the effective duration of lock-up or the return earned while collateral is locked. We use this logic to separate ordinary binary markets from designs that mechanically compress the settlement discount.

Conversion. Some Polymarket mutually exclusive multi-outcome events use a negative-risk (negRisk) architecture [Pol23]. In these events, a multi-outcome question is represented as a collection of binary outcome markets, one for each candidate outcome, and an additional conversion layer links the binary markets within the same event. The key mechanism is that a NO share in one outcome can be converted into YES shares in the other outcomes. In a three-outcome event (A, B, C) , for example, converting $N_A + N_B$ is equivalent to one residual YES on C plus one cash-like par unit. Appendix 9.8 formalizes and proves this payoff identity. The delayed-settlement exposure is therefore concentrated in the smaller residual claim, so we treat negRisk as a distinct design feature and analyze it separately from the non-negRisk baseline.

Collateral yield. A second design mechanism is compensation for locked balances. Polymarket historically operated with largely non-yield-bearing locked collateral, but it later introduced fixed 4% holding rewards for selected long-dated market positions [Shi25, Pol26b]. Kalshi [Kal26b] documents APY paid on collateral balances, with policy rates that can change over time [Kal26a]. Yield support does not eliminate delayed settlement, but it offsets the opportunity cost of waiting and should flatten the observed discount wedge.

3 Settlement-Induced Discounting

3.1 Pricing with Delayed Settlement

$$P_{i,t} = \mathbb{E}_t[X_i] \cdot D(\tau_{i,t}^e), \quad D(\tau) \in (0, 1].$$

Here $X_i \in \{0, 1\}$ is the payoff at settlement, $P_{i,t}$ is the time- t price, and $\tau_{i,t}^e$ is the expected remaining time until the winning claim becomes redeemable. The standard frictionless benchmark corresponds to $D(\tau) = 1$. We use “settlement discounting” as shorthand for the reduced-form pricing wedge induced by delayed redeemability, incorporating opportunity cost, residual settlement risk, liquidity frictions, and capital lock-up.

A convenient parameterization is

$$D(\tau) = \exp(-r_{\text{PM}}(\tau) \tau),$$

where $r_{\text{PM}}(\tau)$ is the reduced-form required return on capital locked in unresolved prediction-market positions, measured in the same time units as τ . It is not a risk-free rate or a pure oracle-delay premium; it is the revealed compensation for holding a delayed, platform-specific settlement claim. The annualized settlement wedge (ASW) recovered below reports this required-return object in annualized terms. Unlike classical prediction-market horizon effects, the discount factor here is directly shaped by settlement architecture, collateral mobility, and convertibility.

The priced horizon is ex ante expected remaining settlement time,

$$\tau_{i,t}^e := \mathbb{E}_t[T_i^{\text{resolution}} - t],$$

where $T_i^{\text{resolution}}$ denotes the priced redeemability timestamp, i.e., T_i^{settle} . In the main structural panel, realized `closeDate` proxies this settlement timestamp. Empirically,

$$\tau_{i,t}^{\text{obs}} = T_i^{\text{resolution}} - t = \tau_{i,t}^e + u_{i,t},$$

where $u_{i,t}$ is forecast error in settlement timing. Under standard non-differential proxy-error conditions, using realized settlement time attenuates estimated horizon effects toward zero rather than mechanically generating a positive horizon gradient. For notational simplicity, subsequent equations write $\tau_{i,t}$.

Settlement–Discount Continuum. The same discount factor links long-horizon below-par pricing and short-horizon post-event carry. At long horizons, the settlement wedge is visible in near-certain prices; near settlement, the level wedge is smaller, but a few basis points can annualize into high carry because the holding period is short. Tick size and discrete order books can prevent the wedge from vanishing completely even after economic uncertainty is gone. These are therefore not separate anomalies, but different points on one maturity-dependent settlement-discount continuum; Appendix 9.3 documents the corresponding short-horizon carry strategies.

3.2 Testable Implications

The framework yields three direct empirical predictions when $D'(\tau) < 0$.

(1) Long-Horizon Near-Certainty Discounts. For near-certain contracts, delayed settlement implies below-par pricing:

$$P_{i,t} = \mathbb{E}_t[X_i]D(\tau_{i,t}) < 1 \quad \text{when } \mathbb{E}_t[X_i] \approx 1, \tau_{i,t} > 0.$$

Among contracts with similar expected payoffs, longer remaining settlement delay implies a lower price. If $\mathbb{E}_t[X_i] \approx \mathbb{E}_t[X_j] \approx 1$ and $\tau_{i,t} > \tau_{j,t}$, then

$$P_{i,t} \approx D(\tau_{i,t}) < D(\tau_{j,t}) \approx P_{j,t}.$$

Interpreting raw prices as probabilities therefore generates a horizon-linked underconfidence pattern, which discount adjustment should attenuate.

(2) Mechanical Price Drift. Conditional on beliefs,

$$\frac{\partial P_{i,t}}{\partial \tau_{i,t}} = \mathbb{E}_t[X_i]D'(\tau_{i,t}) < 0.$$

Equivalently, as calendar time passes and $\tau_{i,t}$ falls, prices drift upward mechanically absent belief revisions.

Table 1: Main analysis samples and measurement choices.

Exercise	Sample	Unit	negRisk	Horizon	Use
Structural panel	Structural panel	Market-day	Mixed	closeDate	Base sample
Baseline frontier	Near-certainty tail	Horizon bin	Excluded	closeDate	Estimate $\hat{D}(\tau)$
Calibration	Non-negRisk tail	Market-day	Excluded	closeDate	Calibration tests
negRisk comparison	Named-outcome negRisk	Weekly	Included	Weeks-to-end	Conversion effect
Kalshi comparison	Cross-platform tail	Weekly	PM baseline	Weeks-to-end	Yield comparison
Carry appendix	Linked trades	Trade	Mixed	Trade-to-settle	Post-event carry

(3) Platform-Design Sensitivity. Platform features operate through $D(\tau)$. Yield-bearing collateral, faster settlement, merging, and conversion paths should raise $D(\tau)$ toward one for a given horizon and compress the wedge; longer lock-up or lower collateral productivity should lower $D(\tau)$ and amplify it. Settlement discounting is therefore endogenous to market architecture.

These predictions motivate the empirical design in the next section: we estimate the settlement-discount frontier, recover ASWs across horizons, test horizon-conditioned calibration, and compare environments with different capital efficiencies.

4 Data and Empirical Design

This section maps the settlement-discount model to the empirical design. We first describe the analysis samples, then define the identification strategy, the near-certainty frontier used to estimate the discount factor, and the calibration adjustment used in the main tests.

4.1 Data Sources and Sample Construction

Data sources. The main empirical setting is Polymarket. We use hourly quote data from the Polymarket Data API and on-chain trade and participant data from the Polygon blockchain [Pol26a]. In the notation above, **endDate** measures T_i^{end} and **closeDate** is the realized proxy for T_i^{settle} in the structural panel. For appendix trade diagnostics, we link UMA oracle timestamps and dispute metadata, with **umaEndDate** as the realized oracle-resolution field. External opportunity-cost benchmarks are AAVE lending rates from Dune [Dun26] and the 2-year constant-maturity Treasury yield from FRED [Boa26].

Sample construction. The raw universe contains all Polymarket markets listed from platform inception through December 31, 2025. The structural sample restricts this universe to markets where delayed settlement can plausibly matter: event duration at least 14 days, cumulative outcome-token volume $\text{volumeNum} \geq 100$, attached CLOB price histories, and non-stale daily quote panels. Within this universe, we focus on markets where it is meaningful to study near-certain pricing: at least one YES or NO midpoint must remain at or above 0.90 for seven consecutive daily snapshots, and markets with subsequent large reversals are excluded. Baseline frontier and calibration exercises exclude negRisk markets to estimate a non-negRisk discount benchmark; negRisk markets re-enter only in the design-comparison exercise. Detailed stage-level sample construction is reported in Appendix 9.1, Table 4.

The structural panel supports frontier estimation, ASW recovery, and horizon-conditioned calibration. Calibration uses one non-negRisk observation per market-day at 00:00 ET, so raw and discount-adjusted diagnostics share the same pricing architecture as the frontier used to estimate $\hat{D}(\tau)$. A separate linked trade dataset contains 19,868,940 trades and supports the short-horizon carry and address-level appendix analyses.

Table 1 summarizes the analysis samples and measurement choices.

4.2 Identification Strategy

The central empirical move is to use persistent upper-tail contracts to recover the discount component in

$$P_{i,t} = \mathbb{E}_t[X_i]D(\tau_{i,t}).$$

Define residual failure probability

$$\delta_{i,t} := 1 - \mathbb{E}_t[X_i] \in [0, 1],$$

so that

$$P_{i,t} = (1 - \delta_{i,t})D(\tau_{i,t}).$$

Rearranging the pricing identity isolates the confound:

$$1 - P_{i,t} = \underbrace{1 - \mathbb{E}_t[X_i]}_{\text{residual uncertainty}} + \underbrace{\mathbb{E}_t[X_i](1 - D(\tau_{i,t}))}_{\text{settlement discount}}.$$

Distance from par therefore combines residual uncertainty and settlement discounting. The main identification threat is horizon-correlated residual uncertainty: a long-horizon near-certain claim may trade below par because it is discounted, because it remains uncertain, or because both forces operate.

Our main estimates use the *high-price near-certainty sample*: claims enter when one side trades persistently in the upper tail. To validate this sample, we separately identify a smaller set of economically resolved markets, where the outcome is already known but oracle finalization and redemption are still pending. In this window, $T_i^{\text{event}} \leq t < T_i^{\text{settle}}$, residual outcome uncertainty is minimal and

$$P_{i,t} \approx D(\tau_{i,t}).$$

Appendix 9.2 reports a compact manual audit of this resolved validation sample.

Identification is exact when $\delta_{i,t} = 0$. In the data, the frontier interpretation rests on four conditions.

1. *Near-certainty*. In the persistent upper tail, residual failure probability is small relative to the price-par wedge.
2. *Frontier selection*. Conditional on horizon, the upper envelope is supported by observations with the least residual uncertainty and smallest idiosyncratic downward frictions.
3. *Proxy horizon*. Realized settlement time is a noisy but informative proxy for the ex ante priced settlement horizon.
4. *No matching residual trend*. After the persistence and reversal screens, frontier-support residual uncertainty is not rising with horizon strongly enough to explain the observed price gradient.

Under these conditions, a high-quantile frontier satisfies

$$P_q(\tau) \approx (1 - \delta_q(\tau))D(\tau),$$

where $\delta_q(\tau)$ is residual uncertainty among frontier-support observations. If $\delta_q(\tau) > 0$, then

$$\hat{D}_q(\tau) = P_q(\tau) \leq D(\tau),$$

and the implied rate is upward biased:

$$\hat{r}_q(\tau) = -\frac{1}{\tau} \log P_q(\tau) = r_{\text{PM}}(\tau) - \frac{1}{\tau} \log(1 - \delta_q(\tau)) \geq r_{\text{PM}}(\tau).$$

The recovered object is therefore a reduced-form settlement wedge. Remaining residual uncertainty biases ASWs upward and makes discount-adjusted calibration an attenuation test rather than exact latent-probability recovery.

4.3 Frontier Estimation

Define the near-certainty frontier $P_q(\tau)$ as the high-quantile envelope of retained upper-tail prices at settlement horizon τ . Using a high quantile rather than the raw maximum limits sensitivity to stale quotes and single-market outliers.

The implied continuously compounded lock-up rate is

$$r_q(\tau) = -\frac{1}{\tau} \log P_q(\tau),$$

with τ measured in days. We report the annualized settlement wedge (ASW) as

$$\text{ASW}_q(\tau) = \exp(365 r_q(\tau)) - 1.$$

The curve $\tau \mapsto \text{ASW}_q(\tau)$ is a reduced-form required-return term structure for locked prediction-market capital.

To track calendar-time evolution, we define

$$\text{ASW}_{\min}(t) = \min_{\tau \in T(t)} \text{ASW}_{q,t}(\tau),$$

where $\text{ASW}_{q,t}(\tau)$ denotes the date- t frontier-implied annualized settlement wedge at horizon τ , and $T(t)$ is the set of observed horizons with sufficient support on date t . Separately, to measure how concentrated the frontier-support tail is across markets at a given settlement horizon, we report

$$N_{\text{eff}}(\tau) = \left(\sum_i w_i(\tau)^2 \right)^{-1},$$

where $w_i(\tau)$ is market i 's share of observations in the frontier-support tail at horizon τ . Operationally, for each horizon bin we compute APYs for all retained observations, select the bottom 5% APY tail, count each market's observations in that tail, and normalize these counts to sum to one. Thus $N_{\text{eff}}(\tau)$ measures the effective number of distinct markets contributing to the low-APY/high-price frontier-support tail. This tail is a nearby support diagnostic for the frontier rather than necessarily the exact set of observations that determines the maximum-price or 0.1-percentile APY line.

4.4 Calibration Design

Calibration compares quoted prices with realized frequencies. Each forecast observation consists of price $P_{i,t}$, horizon $\tau_{i,t}$, and outcome X_i , where $P_{i,t}$ is the midpoint and $\tau_{i,t}$ is time to realized `closeDate`. We use one snapshot per market-day to limit serial dependence and cluster inference by market. Calibration weights are

$$w_i = \sqrt{\max(\text{volume}_i, 1)},$$

where volume_i is outcome-token volume. All calibration exercises use non-negRisk markets so the pricing architecture matches the frontier used to estimate $\hat{D}(\tau)$.

The first calibration object is the raw horizon gradient: whether reliability deteriorates with τ when prices are used directly. Pooling across maturities is not innocuous because, under settlement discounting, the same raw price can correspond to different expected payoffs at different horizons. Because midpoint quotes may be stale or wide in sparse long-horizon markets, the calibration diagnostics should be read as price-signal diagnostics.

The second object is the adjusted horizon gradient. Define

$$\tilde{P}_{i,t} = \min \left\{ 1, \frac{P_{i,t}}{\hat{D}(\tau_{i,t})} \right\},$$

with $\hat{D}(\tau)$ estimated from the non-negRisk near-certainty frontier. The test is whether replacing $P_{i,t}$ with $\tilde{P}_{i,t}$ attenuates the raw horizon gradient. This tests whether the estimated settlement component explains the maturity pattern in apparent miscalibration. We report both in-sample adjustments and event-level cross-fitted specifications, where the frontier is estimated on other events and applied to held-out events.

5 Results

This section establishes four empirical facts. First, near-certain contracts trade below par. Second, the wedge is maturity-dependent and summarized by a frontier-implied ASW term structure. Economically, the ASW curve is the revealed term structure of compensation required to hold capital in delayed-settlement contingent claims. Third, frontier-implied discount curves explain much of the raw long-horizon price-frequency gradient. Fourth, market architecture shifts the wedge as predicted by capital-efficiency mechanisms. We present these facts in turn, then point to possible exploitation strategies in Appendix 9.3.

5.1 Near-Certainty Frontier and Annualized Settlement Wedge

The near-certainty frontier turns below-par prices into a lock-up compensation object. ASW is the reduced-form required return for locked prediction-market capital, bundling opportunity cost, settlement latency, liquidity, platform risk, and residual uncertainty that survives the frontier filters. We document it through the cross-sectional term structure, the lower-tail time series, and external-yield comparisons.

Baseline term structure. As defined in Section 4.3, the baseline term structure maps the near-certainty frontier into ASW by time-to-settlement. Figure 1 plots the minimum ASW at each horizon and the 0.1-percentile ASW as a lower-tail benchmark. Three facts stand out. First, both series are positive throughout. Second, ASW falls sharply from the short end, stabilizes by roughly 20 days, then rises again, with a long-end hump around 230–260 days. Third, the short end is partly constrained by CLOB tick size: a 1.0 ask and 0.999 bid imply a 0.9995 midpoint. The empirical series remain above that bound, so the pattern is not merely mechanical. Because hours or days remain at the short end, tiny near-par discounts annualize into high ASWs. Appendix 9.3 links this wedge to post-event carry strategies with 0.999-style bids held through formal settlement. The curve therefore implies positive, maturity-varying compensation, with elevated annualized rates at the short end and a renewed increase at longer lock-up horizons.

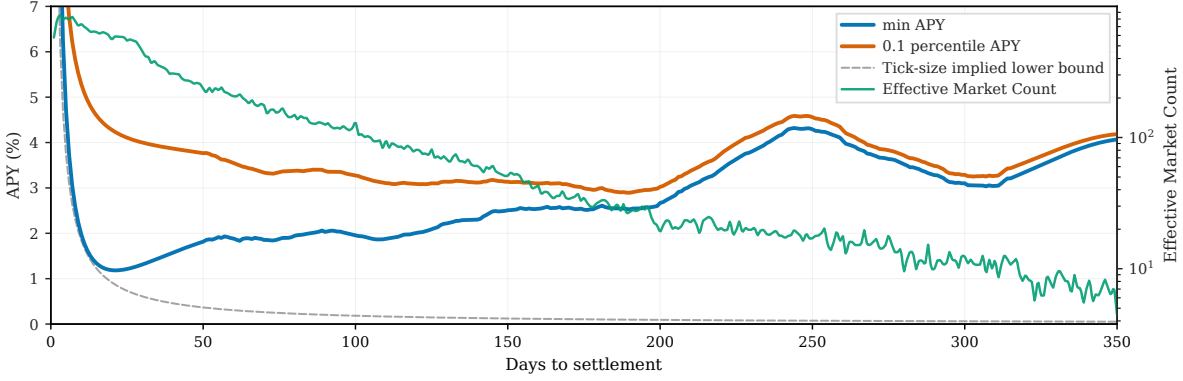


Figure 1: Frontier-implied ASW by time to settlement. The figure plots smoothed minimum and 0.1-percentile ASW across daily contract-level observations. The object is a reduced-form required return on locked prediction-market capital. The dashed line marks the CLOB tick-size lower bound of 0.9995.

Time-series evolution of the platform frontier. The platform frontier is summarized by the daily lower-envelope statistic $ASW_{\min}(t)$, the lower-tail 0.1-percentile ASW, and the horizon-binned support diagnostic $N_{\text{eff}}(\tau)$. Figure 2 shows high lower-tail ASWs early in the sample, when the relevant frontier-support tail is concentrated, consistent with scarce capital. From late 2024 onward, frontier-implied rates compress and the relevant horizon bins show broader effective support: the frontier becomes cheaper and less concentrated. The joint movement of the minimum and lower-tail percentile rules out a single-observation artifact. Polymarket’s later 4% yield support for long-running markets does not produce a comparable upward break. The late sample instead reflects a shift toward shorter horizons, changing the economic content of the daily minimum-rate object.

External benchmark comparison. Figure 3 compares Polymarket’s frontier-implied ASW with external yields. The relationship is state- and maturity-dependent, so the full-period AAVE correlation ($r \approx 0.365$) masks distinct regimes. Before the election, when frontier maturities are longer (median about 144 days), constant-maturity ASWs co-move positively with AAVE (90-day $r = 0.302$, 180-day $r = 0.177$). Around the November 5, 2024 U.S. election, the AAVE link weakens, consistent with broader participation by less crypto-native traders. Between the election and Polymarket’s 4% yield program, ASWs compress and benchmark linkage weakens as support thickens. Afterward, frontier

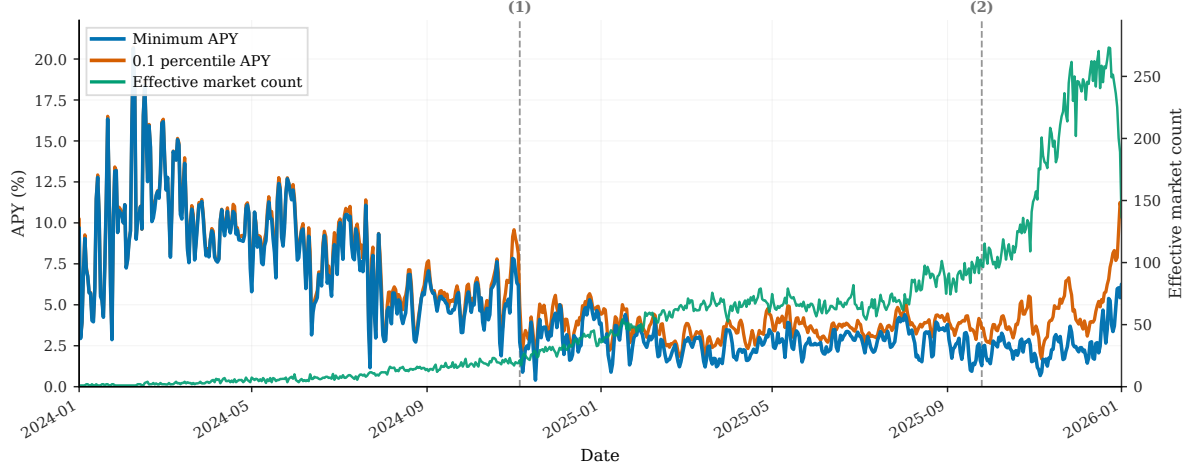


Figure 2: Time-series evolution of lower-tail frontier-implied ASWs with horizon-binned effective support. The figure shows smoothed minimum ASW, 0.1-percentile ASW, and the observation-weighted effective market count $N_{\text{eff}}(\tau)$ for the bottom 5% APY frontier-support tail. Because $N_{\text{eff}}(\tau)$ is computed by time-to-settlement bin rather than calendar date, it should be read as a concentration diagnostic for the displayed frontier-support tail, not as a pure time-series count of unique markets. Solid vertical lines mark (1) the November 5, 2024 U.S. election and (2) the introduction of Polymarket's 4% yield program.

maturity shifts to the short end (median about 38.5 days), so apparent sign flips mainly reflect composition and microstructure, not a reversal in long-horizon capital pricing. Treasury correlations stay weak, and lead-lag tests show no significant directional relationship with AAVE yields. Appendix 9.4 shows the corresponding unsmoothed daily scatter.

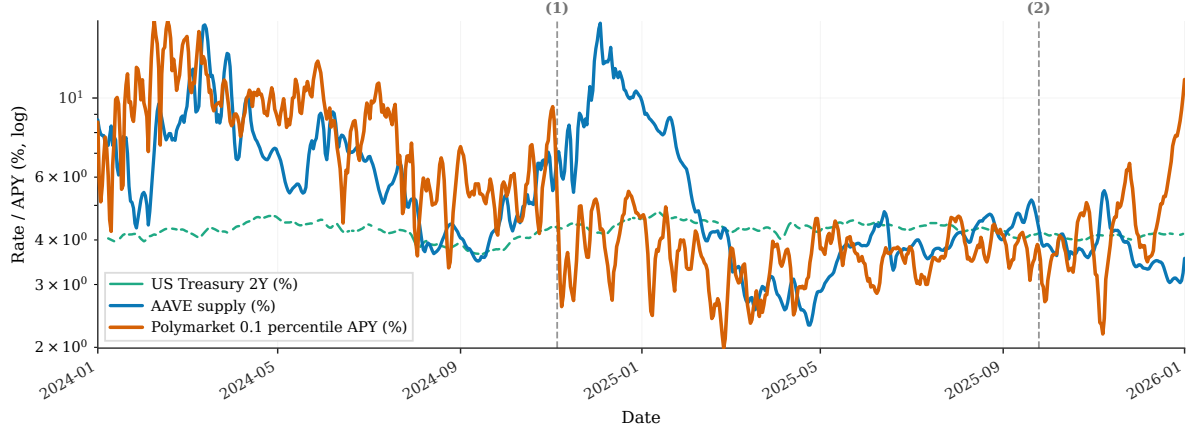


Figure 3: Comparison of external benchmark yields with Polymarket's lower-tail frontier-implied ASW. The figure plots smoothed AAVE supply rates, 2-year U.S. Treasury yields, and Polymarket's 0.1-percentile ASW.

These results establish a positive, horizon-varying settlement discount in the data. The next subsection asks how much of the raw calibration-horizon pattern survives after discount adjustment.

5.2 Horizon-Conditioned Price-Signal Distortion

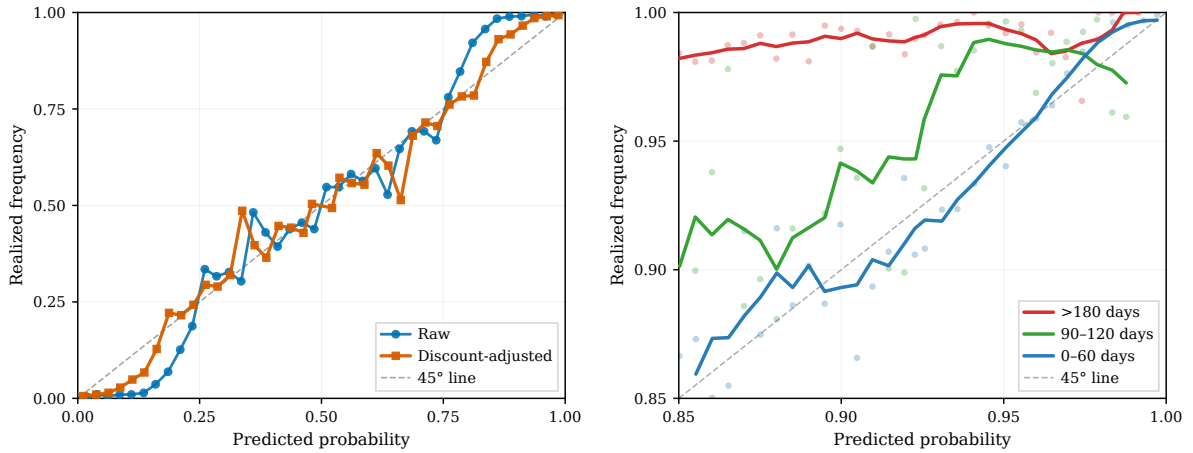
Using the settlement discount above and the design in Section 4.4, we test whether the raw price-frequency gap steepens with settlement horizon and attenuates after discount adjustment. All estimates use the non-negRisk structural sample, daily market-level observations, and market-clustered inference.

Table 2: Calibration summary for raw and discount-adjusted prices across disjoint settlement-horizon buckets. Brier score measures overall forecast error; MSE^a measures deviation from the reliability diagonal and is the primary calibration diagnostic. Percentage improvements are relative to the raw metric.

Horizon	N	Brier (raw)	Brier (adj)	% imp.	MSE ^a (raw)	MSE ^a (adj)	% imp.
0–30 days	194406	0.032310	0.032259	0.16%	0.003102	0.003066	1.15%
30–60 days	109123	0.063199	0.063031	0.27%	0.004041	0.003831	5.20%
60–120 days	111181	0.084946	0.084655	0.34%	0.003255	0.002865	11.99%
120–180 days	50769	0.085622	0.084938	0.80%	0.005903	0.005183	12.18%
> 180 days	41933	0.106967	0.106310	0.61%	0.004916	0.003932	20.02%

^a MSE denotes weighted calibration-gap mean squared error, i.e. squared deviations from the reliability diagonal.

Reliability Diagnostics. Figure 4 separates raw and adjusted reliability. In panel (b), raw longer-horizon contracts sit progressively above the 45-degree line, indicating stronger underprediction as time-to-settlement rises. In panel (a), discount-adjusted long-horizon reliability based on $\hat{D}(\tau)$ shifts materially toward the diagonal. The adjustment absorbs much of the long-horizon price-signal distortion, especially where the raw settlement wedge is largest.



(a) Raw and discount-adjusted reliability for contracts with settlement horizons above 180 days. Observations are volume-weighted; adjusted prices use the date-varying envelope discount $\hat{D}(\tau)$.

(b) Horizon-conditioned reliability for raw near-certainty prices across settlement-horizon buckets. Observations are volume-weighted; curves are rolling-mean smooths.

Figure 4: Calibration diagnostics in the near-certainty region using the same non-negRisk dataset, daily-midpoint sampling, and volume weighting. Panel (a) shows that discount adjustment moves long-horizon reliability toward the 45-degree line. Panel (b) shows that raw reliability deteriorates with settlement horizon, matching settlement-induced discounting.

Table 2 gives the horizon-bucket summary. Raw Brier score deteriorates monotonically with horizon (0.0323 in the 0–30 day bucket versus 0.1070 beyond 180 days), so raw prices appear less accurate at longer maturities. Discount adjustment improves Brier score only modestly, because Brier score mixes calibration error with residual outcome uncertainty. The weighted calibration-gap MSE more directly isolates deviations from the reliability diagonal: improvements rise from 1.15% at short horizons to 20.02% in the longest bucket. Appendix 9.5 visualizes this pattern. The key diagnostic is the maturity gradient: adjustment matters most where capital remains locked longest.

Regression Evidence. As a formal complement, we estimate the horizon gradient in the pricing wedge. The goal is to summarize the graphical pattern in one coefficient and test whether it survives

Table 3: Tail regression of the pricing wedge on time-to-settlement in the near-certainty sample $P \geq 0.9$. P-values are clustered by market. Adjusted specifications use $\tilde{P}_{i,t}^{(q)}$ constructed from date-varying frontier-implied discount curves estimated from the minimum, 0.1-percentile, and 0.5-percentile near-certainty envelopes. Mean implied ASW reports the average ASW along the corresponding frontier curve. The sample contains 154,298 observations across 4,273 market clusters.

Discount curve	Mean ASW	$\hat{\beta}$	Clustered p-value	Slope reduction
Raw (no adjustment)	—	0.00016780	3.75×10^{-7}	—
Minimum ASW frontier	3.06%	0.00008757	0.0079	47.8%
0.1-percentile frontier	4.36%	0.00006686	0.0425	60.2%
0.5-percentile frontier	6.89%	0.00002076	0.5285	87.6%

discount adjustment using frontier-implied curves:

$$X_i - P_{i,t} = \alpha + \beta \tau_{i,t} + \varepsilon_{i,t},$$

$$X_i - \tilde{P}_{i,t}^{(q)} = \alpha_q + \beta_q \tau_{i,t} + \varepsilon_{i,t}, \quad \tilde{P}_{i,t}^{(q)} := \min \left\{ 1, \frac{P_{i,t}}{\hat{D}_q(\tau_{i,t})} \right\},$$

for frontier choices $q \in \{\min, 0.1, 0.5\}$. All specifications use the near-certainty sample ($P \geq 0.9$) and market-clustered standard errors. Here $\tau_{i,t}$ proxies for the latent expected horizon priced ex ante, so non-differential proxy noise attenuates $\hat{\beta}$ toward zero. Table 3 shows a positive, highly significant raw horizon gradient ($\hat{\beta} = 0.00016780$, $p = 3.75 \times 10^{-7}$). The date-varying frontier-implied curves then attenuate it monotonically. The minimum-ASW frontier (mean ASW 3.06%) lowers the slope to 0.00008757 (47.8%, $p = 0.0079$); the 0.1-percentile frontier (4.36%) lowers it to 0.00006686 (60.2%, $p = 0.0425$); and the conservative 0.5-percentile frontier (6.89%) lowers it to 0.00002076 (87.6%, $p = 0.5285$), statistically indistinguishable from zero. Thus discount adjustment reduces the horizon gradient by roughly 48–88%, depending on the lower-tail frontier; Appendix 9.6 provides the corresponding visualization.

Event-level 5-fold cross-fitting in Appendix 9.7 tests whether the frontier is portable across event sets: each held-out fold is adjusted using frontiers estimated from other events. Held-out slope reductions remain large (56.4–92.9%), so attenuation does not weaken when adjusted events are excluded from frontier estimation.

Calibration and regression evidence point to the same conclusion: the horizon effect is economically consistent with discounting. Frontier-implied curves, not chosen to eliminate the slope, reduce the raw maturity gradient by roughly 48–88%; event-level cross-fitting preserves the ordering. A substantial share of the long-horizon price–frequency gap therefore reflects settlement-induced pricing of locked capital, not only belief error.

5.3 Design-Driven Comparative Statics

Market design shapes the level and slope of settlement discounting. Convertibility and collateral yield determine how quickly capital can be recycled and how costly waiting is. We therefore treat design differences as comparative statics of the pricing equation, not shifts in trader rationality.

5.3.1 NegRisk Conversion and Discount Compression

NegRisk provides the cleanest design comparative static. The key point is that a basket of NO tokens can be converted into a deterministic cash-like component plus the complementary YES basket. For any subset S of m outcomes in an n -outcome event, Appendix 9.8 derives the payoff identity

$$\sum_{k \in S} N_k \equiv (m - 1) \cdot \mathbf{1} + \sum_{j \notin S} Y_j.$$

This payoff identity underlies Polymarket’s NegRiskAdapter. The deployed adapter exposes the NO-basket-to-complementary-YES conversion direction; at the payoff level, the identity says that the NO basket is economically equivalent to $(m - 1)$ par units plus only the residual complementary YES

exposure. Thus conversion concentrates delayed-settlement risk in the residual leg rather than in the full NO basket. Let $V_S(t)$ denote the time- t value of this NO basket. Since the deterministic component is valued at par,

$$V_S(t) := \sum_{k \in S} P(N_k, t) = (m - 1) + \sum_{j \notin S} P(Y_j, t).$$

Under the reduced-form model $P(Y_j, t) = p_j(t)D(\tau)$, with $\sum_{j=1}^n p_j(t) = 1$, this becomes

$$V_S(t) = (m - 1) + (1 - p_S(t))D(\tau), \quad p_S(t) := \sum_{k \in S} p_k(t).$$

The empirically relevant case is a tail NO basket: the outcomes in S have been pushed close to elimination, so their combined YES probability satisfies $p_S(t) \approx 0$. Then $V_S(t) \approx (m - 1) + D(\tau)$. If the m NO tokens are similarly priced, their average price $\bar{P}_N(\tau; m) := m^{-1} \sum_{k \in S} P(N_k, t)$ is

$$\bar{P}_N(\tau; m) \approx \frac{(m - 1) + D(\tau)}{m} = 1 - \frac{1 - D(\tau)}{m}.$$

Thus conversion divides the standard binary settlement wedge $1 - D(\tau)$ across the m nettable NO legs. In the largest nettable basket of an n -outcome event, $m = n - 1$, leaving only one residual YES claim and giving the near-certain negRisk benchmark

$$\bar{P}_N(\tau; n - 1) \approx 1 - \frac{1 - D(\tau)}{n - 1}.$$

Appendix 9.8 gives the full payoff derivation and Appendix 9.9 derives fee-adjusted bounds.

Empirical results. Figure 5 compares the empirical upper envelope of near-certain prices with conversion-implied negRisk support lines. The envelope uses weekly 99.9th-percentile raw-price frontiers by weeks before end. Three patterns match the theory. First, negRisk markets stay closer to par than non-negRisk markets across most horizons. Second, within negRisk, events with more linked outcomes (e.g., $n = 6, 7, 8$) lie above smaller events (e.g., $n = 3, 4, 5$), consistent with stronger netting. Third, the ordering is clearest at medium and long horizons, where the wedge is largest; near maturity, all series converge toward par. Far-tail subgroup lines are noisier as support thins, but the central-horizon ordering remains stable.

The empirical envelope therefore reproduces the conversion-implied ordering in Figure 5. Economically, negRisk turns delayed redemption into a mostly cash-like claim, compressing the settlement wedge by construction; compression strengthens with event size. The benchmark treats outcome sets as static and tail probabilities as very small; in the appendix we consider augmented negRisk-events, non-negligible residual probabilities, conversion fees, and taker fees.

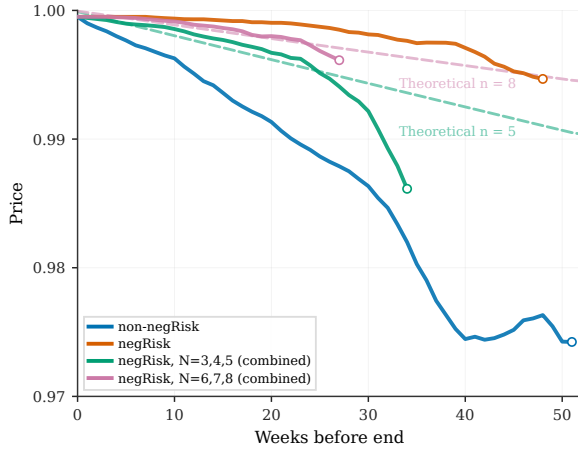


Figure 5: Empirical 99.9th-percentile price frontier by weeks before end. Solid lines show trailing moving averages of weekly horizon aggregates and are truncated where support falls below 50 markets. Dashed lines show conversion-implied benchmarks $\bar{P}_N(\tau; n)$ for $n = 5$ and $n = 8$ under a constant 4% discount factor.

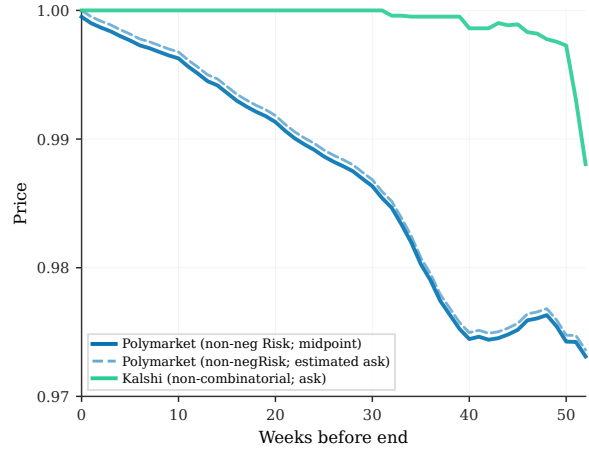


Figure 6: Empirical 99.9th-percentile price frontier by weeks before end. The zoomed near-par scale highlights a cross-platform comparative static: Kalshi remains closer to 1 over most horizons, consistent with flatter effective lock-up premia.

5.3.2 Yield-Bearing Collateral and Flatter Settlement Premia

Yield-bearing collateral provides a second design comparative static. Rather than shortening lock-up through conversion, it flattens $\hat{D}(\tau)$ itself: if collateral remains productive while awaiting settlement, delay is less costly. Near-certain contracts should trade closer to par and imply a flatter ASW term structure.

Figure 6 compares non-negRisk Polymarket contracts with Kalshi. Visually, Kalshi’s price frontier stays closer to par and is flatter across maturities than Polymarket’s. Through $P_{i,t} = \mathbb{E}_t[X_i]D(\tau_{i,t})$, that flatter near-par price profile implies a flatter ASW term structure, which is the comparative static predicted when collateral earns yield.

We interpret the figure as a cross-platform comparative static, not an isolated treatment effect: Polymarket and Kalshi differ in users, fees, composition, liquidity, collateral rules, and resolution. Read with the cleaner negRisk result, however, the comparison reinforces the design implication: convertibility and collateral yield compress the wedge through distinct capital-efficiency channels.

5.4 Trading Exploitation Implications

The same term-structure logic has trading strategy implications, documented descriptively in Appendix 9.3. At the short end, tiny pre-settlement discounts can annualize into large *ex post* carry because only hours or days remain; at the long end, near-certain contracts can temporarily deviate from the platform ASW frontier. Both are retrospective exploitations of the same settlement-discount.

6 Discussion

Core interpretation. Our results suggest that prediction markets should be analyzed as joint information-aggregation and capital-allocation systems. A collateralized claim is not only a forecast about an event; it is also a claim on a delayed, platform-specific dollar. When settlement is delayed and collateral is locked, prices embed a funding component tied to time, outside opportunities, liquidity, oracle risk, and market design. The standard price-as-probability benchmark remains useful, but it is incomplete whenever the discount factor varies across maturities or architectures. The central implication is that price quality is endogenous to capital efficiency.

Implications for calibration. Calibration exercises confound belief quality and funding frictions unless settlement horizon is controlled for. A long-horizon claim can trade below its eventual payoff not because traders doubt the outcome, but because the payoff is costly to hold until formal redemption. This mechanism gives a direct interpretation to the horizon gradients documented above: discount adjustment moves long-maturity near-certainty prices closer to realized outcomes because part of the raw price–frequency gap is a settlement wedge.

Settlement discounting complements rather than replaces behavioral accounts. Favorite–longshot bias and other demand-side forces may still affect prices, but they operate in an environment where capital is costly to immobilize. The mechanism also gives a concise way to interpret some distributional patterns. If near-certain claims are discounted, the corresponding long-shot side can be mechanically expensive relative to a discount-adjusted benchmark, and less sophisticated traders may still buy those mechanically overvalued long-shot positions. This is consistent with, but does not by itself identify, the profit concentration and long-shot losses documented by Akey et al. [AGHM26]. Conversely, sophisticated participants may earn carry-like returns when they can identify claims that are economically safe but formally unsettled, subject to liquidity, fees, and resolution-rule risk.

Capital efficiency as a market-design variable. The most direct design implication is that platforms can improve pricing by lowering locked collateral amounts. Conversion is one lever: by shortening effective lock-up, negRisk compresses the settlement wedge, especially in large outcome sets and when conversion frictions are low [Pol23].

Yield-bearing collateral attacks the same wedge from the other side by reducing the opportunity cost of lock-up. Kalshi’s APY policy and Polymarket’s selected holding-rewards program are therefore naturally read as market-design responses to long-horizon pricing deterioration [Kal26a, Pol26b]. The cross-platform evidence is only suggestive, because platforms differ along many dimensions, but it is directionally consistent with the model: when collateral or open positions earn yield, the implied ASW curve should flatten. For decentralized CTF-style markets, non-rebasing yield-bearing collateral provides a clean implementation route because it preserves a stable accounting unit while allowing yield to accrue in the background; TrueO’s TYD is one example of this approach [Tru26a, Tru26b].

Polymarket’s targeted 4% rewards also have a possible unintended effect. If traders can earn the reward by holding both YES and NO in eligible long-running markets, the resulting complete-set position becomes a lower-risk platform outside option than marginal exposure to a near-certain frontier claim. Rational traders may then require non-rewarded settlement claims to clear a higher implied ASW, widening rather than compressing the platform-wide wedge. We do not yet see this clearly in recent data, but the mechanism makes targeted yield support potentially counterproductive.

The broader tradeoff is that yield support is visible and easy to understand, but can subsidize positions whose settlement wedge is already small. This is especially relevant for Polymarket negRisk markets: because conversion already recycles capital, subsidizing them is not necessary if the goal is simply to offset the cost-of-capital effect. Support is more valuable where collateral remains fully locked. Conversion is more targeted because it compresses the wedge by construction, but its effectiveness depends on fees. Taker fees and conversion fees widen convergence bands, while low-friction conversion makes the synthetic collateral channel stronger. From a protocol-design perspective, the relevant question is how cheaply the market can recycle capital while preserving credible settlement.

Limits of interpretation. The empirical objects in this paper are reduced-form wedges, not a full structural decomposition of required returns. Persistent upper-tail prices minimize residual uncertainty, but do not eliminate it; some near-certain claims may still contain event risk. Midpoint prices may be stale or non-executable in thin markets, especially at longer horizons. Realized settlement time is an observable proxy for the expected settlement horizon priced by traders at the time of trade. Finally, cross-platform comparisons should be read as comparative statics rather than causal treatment effects, since platforms differ in users, fees, regulation, collateral rules, liquidity, and resolution procedures.

The broader conclusion is simple: prediction markets are pricing systems under collateral constraints. Their prices aggregate information, but they do so through contracts that must be funded, held, and eventually redeemed. Accounting for that financing environment is therefore necessary for calibration, cross-platform comparison, and market-design evaluation.

7 Related Work

Our paper sits at the intersection of prediction-market interpretation, cost-of-capital pricing, and collateral-based valuation. The standard price-as-probability mapping remains our benchmark, but delayed settlement and locked collateral make the observed price a discounted expected payoff rather than a pure probability signal.

Wolfers and Zitzewitz [WZ04] establish the canonical frictionless benchmark: under risk-neutral, or approximately risk-neutral, conditions, winner-take-all prices can be interpreted as market-implied probabilities. Manski [Man06] provides a different critique, showing that heterogeneous beliefs and budget constraints can prevent market price from identifying mean belief. Both papers clarify the belief-to-price mapping.

The closest direct prediction-market predecessor is Page and Clemen [PC13]. They document calibration distortions and horizon effects with explicit attention to time to expiration, opportunity costs, and participation/no-trade regions. Their Intrade setting makes it difficult to separate event uncertainty, expiration horizon, platform costs, and settlement/redeemability frictions. By contrast, realized settlement timestamps and persistent near-certainty frontiers allow us to recover a discount curve $\hat{D}(\tau)$ more directly and interpret the remaining horizon pattern as a settlement wedge.

Grant, Johnstone, and Kwon [GJK19] provide the closest cost-of-capital lens within prediction-market theory. They model prediction contracts as risky assets with required returns, producing a belief- and investor-specific discount that can rationalize favorite-longshot distortions. Our mechanism is complementary but different: the wedge we study is infrastructure- and horizon-specific, generated by delayed redemption and collateral lock-up. Relative to their framework, our contribution is to identify this settlement wedge in field data, measure its term structure, and show how market design compresses or amplifies it.

Outside prediction markets, collateralized-derivatives research provides a useful finance anchor. Johannes and Sundareshan [JS07] show that collateralization and mark-to-market conventions alter discounting and generate a time-varying cost-of-collateral term structure, while Fujii, Shimada, and Takahashi [FST10] show that collateral remuneration and form are not valuation-neutral. The key analogy is empirical as well as conceptual: a latent collateral-cost component can be recovered from market prices and is economically meaningful. We apply this financing logic to prediction markets by recovering a latent settlement wedge from near-certain prices with realized oracle-finalization horizons.

Finally, recent simulation evidence by Maresca [Mar26] provides complementary support for the same mechanism. In a controlled market experiment, Maresca studies whether remunerating open positions mitigates long-horizon distortions and finds that interest-bearing positions improve participation and price accuracy. Our contribution is different in both object and setting. Rather than estimating the effect of paying interest in a designed environment, we use field data to recover the settlement discount embedded in observed prices, trace its horizon profile using realized settlement time, and compare market architectures with different capital-efficiency properties. Maresca therefore provides intervention evidence on the effect of remunerating locked capital, whereas we provide a valuation-based interpretation of prices under delayed settlement. The similarity in direction between Maresca’s roughly 83% mitigation estimate and our finding that a conservative 0.5-percentile frontier adjustment shrinks the near-certainty horizon gradient by about 88% is suggestive, but it should not be read as an replication, since the underlying estimands differ.

8 Conclusion

This paper identifies settlement-induced discounting as a measurable pricing component in collateralized prediction markets. The standard price-as-probability benchmark remains useful, but in these markets observed prices are discounted probabilities rather than frictionless probabilities: they are prices of delayed contingent claims. When formal redemption is delayed, prices combine beliefs about outcomes with settlement delay, locked-capital opportunity cost, liquidity demand, and residual platform or oracle risk. This component helps account for long-horizon underpricing, post-event carry, and maturity-dependent calibration patterns without treating them as purely informational or behavioral errors.

Empirically, we recover settlement-implied discount curves from persistent near-certain contracts and convert them into annualized settlement wedges. Discount adjustment reduces the near-certainty

horizon gradient by roughly 48–88%, showing that a large share of the raw maturity pattern reflects a priced settlement wedge rather than forecast error alone. The horizon structure also links the long and short ends of the market. Distant near-certain claims trade below par; near resolution, small residual discounts can annualize into large carry.

Market design is first-order for this pricing component. NegRisk conversion and yield-bearing collateral compress the wedge through distinct capital-efficiency channels: conversion recovers part of the position as synthetic collateral, while yield reduces the opportunity cost of lock-up. The cross-design evidence is consistent with these comparative statics, even though it should not be read as a standalone treatment effect.

For researchers and platform designers, the implication is straightforward: evaluation frameworks should separate information aggregation from funding frictions. Apparent calibration errors can partly reflect financial architecture rather than bad forecasting. Calibration and efficiency claims should be maturity-conditioned, and interventions should target capital efficiency directly—through collateral yield, conversion-friendly market structure, and microstructure choices near par. The broader contribution is an empirical framework for measuring how settlement mechanics and protocol architecture enter prices in modern prediction-market infrastructure.

Acknowledgements

The complete dataset, including all transient transformations and intermediate processing steps, as well as the source code used for the analysis, will be published after publication.

9 APPENDIX

9.1 Sample Construction and Coverage

Table 4 summarizes how the empirical sample is constructed from the full Polymarket event universe. The table separates the broad eligibility screen from the near-certainty screen used for identification.

Table 4: Stage-level sample construction and coverage.

Stage	Restriction	Events	Markets	negRisk markets	Price obs.	Market-days
Full universe	Full Polymarket crawl	141,848 (100.0%)	323,342 (100.0%)	98,186 (30.4%)	n/a	n/a
Eligible CLOB sample	Volume > 100, > 2 weeks, CLOB history	10,608 (7.5%)	62,291 (19.3%)	41,439 (66.5%)	4,647,145	1,473,282
Near-certainty sample	Seven-day upper-tail midpoint screen	4,483 (3.2%)	36,349 (11.2%)	27,601 (75.9%)	3,712,775	1,181,226

The full universe is collected from active, closed, and archived Gamma event documents. The eligible CLOB sample is a coverage and activity screen: it keeps markets with positive volume, open intervals longer than two weeks, and usable order-book histories. The near-certainty sample then applies the identification screen, requiring a YES or NO midpoint of at least 0.90 for seven consecutive daily snapshots from hourly quote data. Price observations are valid hourly YES/NO midpoint observations; market-days are distinct (event, market, UTC date) observations with positive remaining settlement horizon. Event and market percentages are relative to the full universe, and negRisk percentages are row shares; the filters raise the negRisk share from 30.4% to 75.9%, consistent with convertible structures being common among longer-duration markets where capital lock-up is salient.

9.2 Manual Validation of Frontier-Support Observations

The key identification concern is horizon-correlated residual uncertainty: a long-horizon upper-tail contract may trade below par because the outcome is not fully certain, not only because settlement is delayed. The main analysis addresses this through persistent frontier selection and economically resolved but unsettled windows. Table 5 below is therefore a plausibility check: it shows that representative frontier-support observations involve externally resolvable states where residual event risk appears small relative to the observed discount.

Table 5: Manual validation examples for near-certainty frontier-support observations.

Market	Class	Validation rationale
U.S. CPI $\geq 8.1\%$ / 10.1% in 2025	Objective threshold	External BLS threshold; realized inflation was far below the cutoff, so residual uncertainty is small relative to the observed discount.
Is Earth flat?	Low-plausibility	Externally resolvable factual proposition; residual uncertainty is plausibly negligible relative to the observed discount.
Will Jesus Christ return in 2025?	Low-plausibility	Date-specific low-plausibility proposition; useful as a stress-case diagnostic for non-informational demand at the upper frontier.
U.S. defaults on debt by 2027?	Benchmarkable tail event	Externally assessable sovereign credit event; CDS spreads provide an outside benchmark for residual default risk. ²

9.3 Supplementary Strategy Evidence

This appendix focuses on short-horizon liquidity-demand carry. Once the real-world event is economically resolved, a winning claim is close to a delayed \$1 payoff. Holders who want immediate balance-sheet release may sell slightly below redemption value, while buyers earn carry for warehousing the remaining oracle and settlement latency. The price discount can be only a few basis points, but annualization over hours or days can be large; the spread need not vanish instantly because settlement is non-instantaneous, inventory is balance-sheet intensive, and near-par quoting is discretized by tick size.

The object below is not buying high-probability contracts quoted at 0.999: it is a settlement-liquidity trade in economically resolved claims. The next segment defines that screen.

9.3.1 Operational Definition and Screen

We define *bonding trades* narrowly as post-event settlement-liquidity trades: the economic outcome is no longer in doubt, oracle finalization and redemption are still pending, and the orderbook contains bids for the winning claim at 0.999 but no asks below par. This is not the common Polymarket use of “bonding” for buying any high-probability share.

Operationally, we fetch the latest 1,500 trades for each market in the full Polymarket universe and search for post-event regions with closed status, no active dispute, settlement after event end, and a midpoint near 0.9995. Such a midpoint corresponds to bid-side liquidity around 0.999 and an ask effectively at 1.000: traders cannot buy the claim cheaply from the book, but holders can sell certain shares into available bids for immediate cash. We retain fills consistent with this sell-into-bid region and compute carry from the discount to redemption value over the observed time to settlement.

9.3.2 Market Timeline and Post-Event Carry Window

Figure 7 presents a case study of Polymarket’s event “Will Russia capture Kostyantynivka by August 31?”³ The contract asked whether Russia would capture the Kostyantynivka railroad station by August 31, 2025 at 11:59 PM ET. It resolved as NO through Polygon UMA Optimistic Oracle V2⁴ after a September 1, 2025 proposal and settlement later the same morning.

The figure aligns trades with the relevant information update, market close, and oracle settlement. The shaded region is the case-study carry window: it starts at the final transition to a midpoint of 0.9995 and ends at oracle settlement. Before this window, trades still reflect ISW battlefield updates and two-sided price discovery. Inside it, outcome uncertainty is effectively gone and the book is bid near 0.999 with asks at par: holders can sell certain NO shares into passive bids for immediate cash, while liquidity providers warehouse the remaining settlement latency.

²For example, U.S. sovereign CDS spreads during the sample period were on the order of roughly 15 basis points, providing an external market-based proxy for residual default risk [Wor26].

³Polymarket market page: <https://polymarket.com/event/will-russia-capture-kostyantynivka-by>. The market used the ISW Ukraine map as the primary resolution source, with DeepStateMap or credible reporting as fallbacks if needed. A bulletin-board clarification required qualifying ISW shading to persist through the next full ISW daily update cycle and excluded temporary map glitches and infiltration-only shading.

⁴UMA settled oracle record: <https://oracle.uma.xyz/settled?project=Polymarket&transactionHash=0x3b7eef4440ae5a1c6d7c87a02d32fe460aee93849666b345c33582b97921921e&eventIndex=9>.

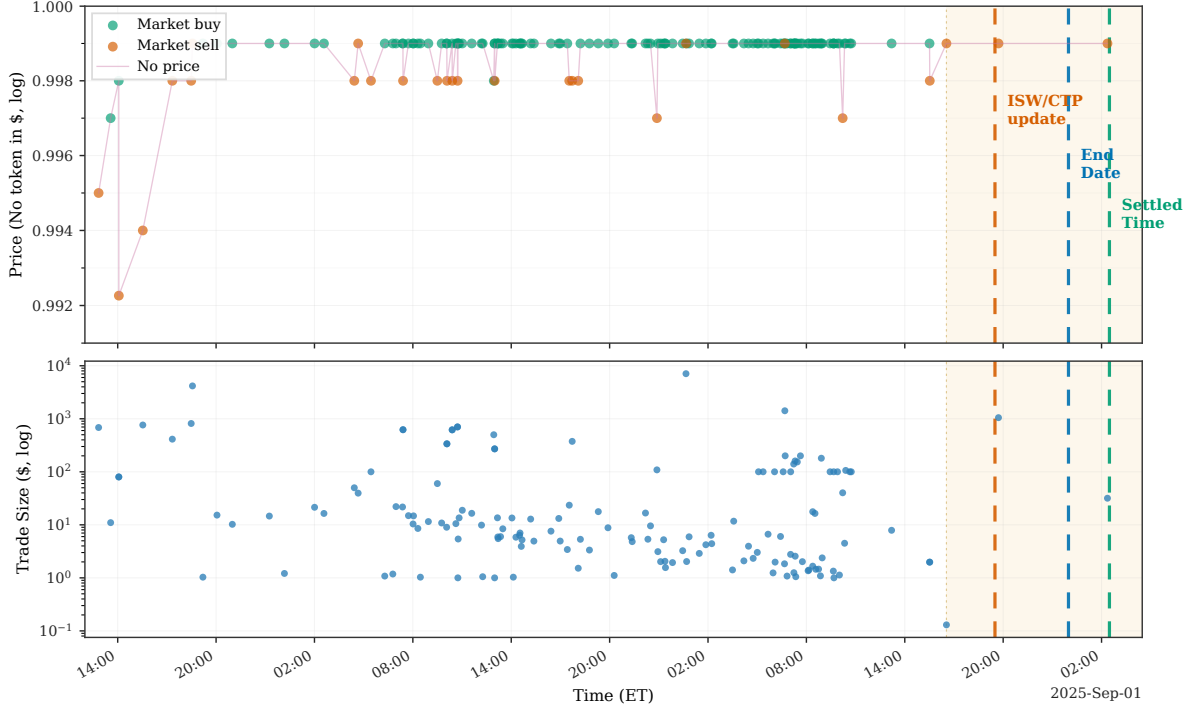


Figure 7: Post-event trading in the Polymarket contract “Will Russia capture Kostyantynivka by August 31?” The upper panel plots the NO-token price around the relevant milestones; the lower panel plots trade size on a log scale.

9.3.3 Cross-Sectional Bonding Evidence

The aggregate evidence characterizes the delay between the scheduled EndDate and official resolution. This timing is not known ex ante: markets can be economically resolved while claims remain non-redeemable, because after EndDate they must still wait for oracle finalization. Under the undisputed optimistic-oracle path, resolution takes two hours, while disputes can extend the interval. Figure 8 shows the distribution of official resolution times relative to EndDate. Most markets resolve on the same day as their EndDate, indicating that the typical post-EndDate wait is short even though the exact resolution time is uncertain beforehand.

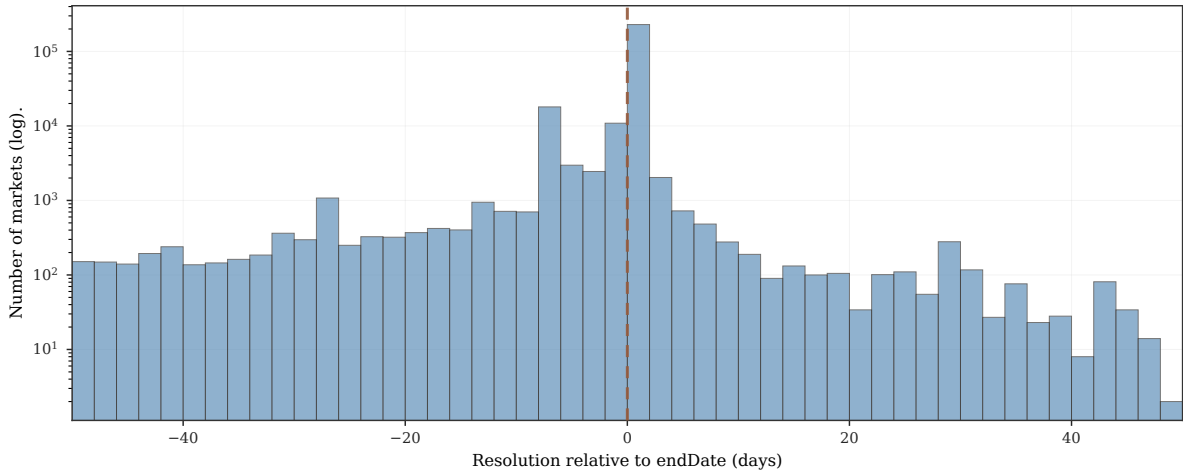


Figure 8: Distribution of remaining settlement horizons for identified bonding trades. The figure shows when short-horizon settlement-liquidity provision occurs in the screened activity.

Figure 9 shows the aggregate scale of settlement-liquidity provision in the short-horizon carry

screen. It reports both the open interest associated with the screened activity and the profits earned by the identified actors over time.

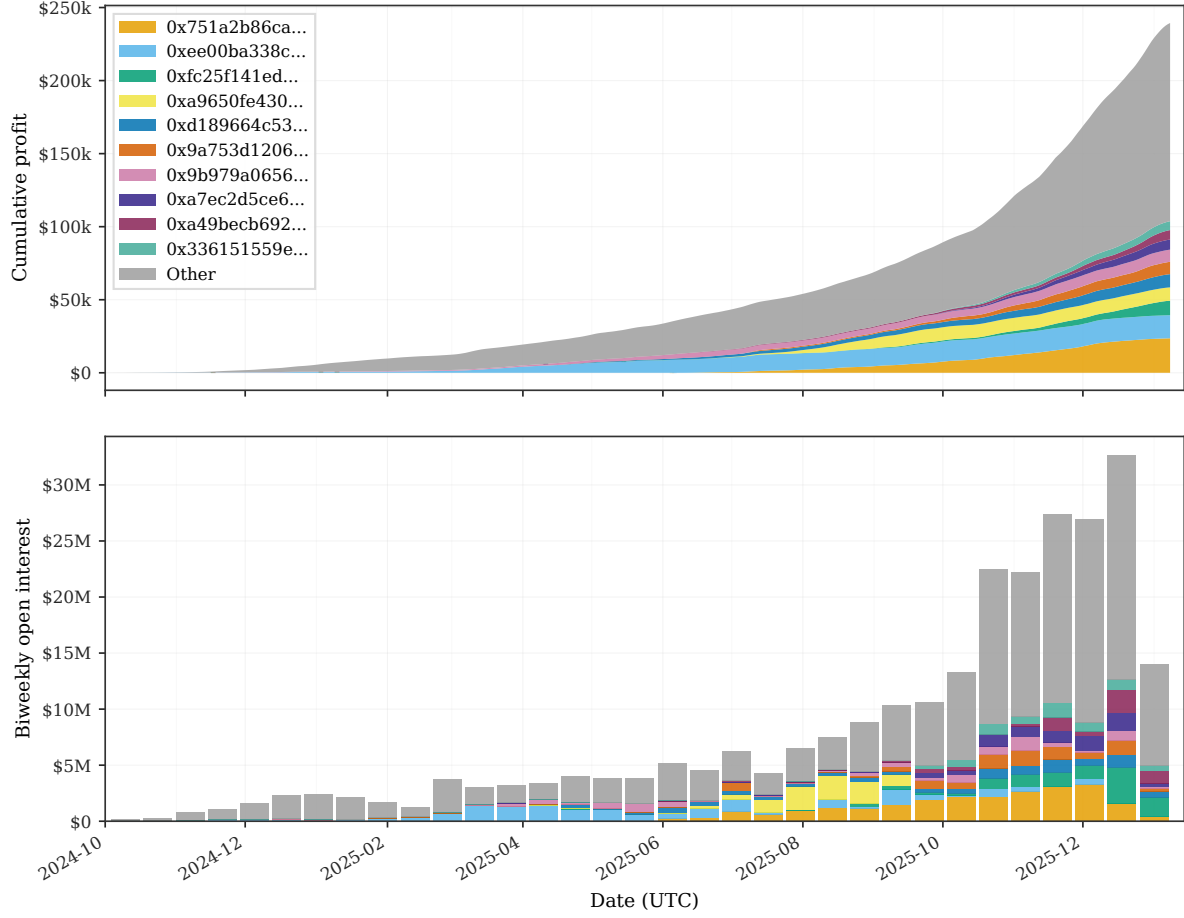


Figure 9: Settlement-liquidity provision in the short-horizon carry screen. The top panel shows stacked carry-screen profit; the lower panel shows bi-weekly open interest in the screened activity.

Table 6 provides the corresponding actor-level summary for the largest identified liquidity providers in the short-horizon carry screen, sorted by total carry profit. The large annualized figures arise mechanically from scaling very small nominal spreads over holding periods often measured in hours.

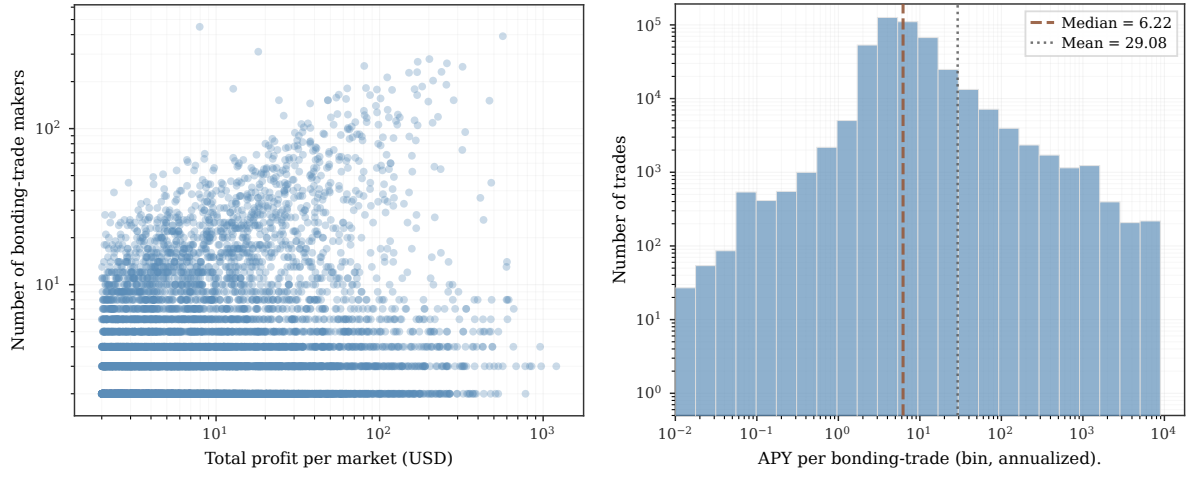
Table 6: Descriptive, retrospective summary of top short-horizon settlement-liquidity providers.

Address	P&L (\$)	# Trades	Median Profit (%) ^a	Median Profit (\$)	Median hold (hours)
0x751a	23616.35	34994	531.67	0.040000	1.0644
0xee00	15805.16	15297	428.85	0.068490	2.0044
0xfc25	9994.85	9649	527.45	0.039740	1.5856
0xa965	9113.03	24440	750.11	0.047515	0.7786
0xd189	9023.79	22225	565.67	0.043040	1.0822
0x9a75	8501.08	9613	511.37	0.042000	1.7950
0x9b97	8345.02	10340	441.72	0.058820	1.9799
0xa7ec	6924.55	11256	446.03	0.040000	1.5369
0xa49b	6537.72	6089	538.01	0.041810	1.6033
0x3361	6092.94	12988	483.59	0.040350	1.5911

^a Percentage column reports holding-time-adjusted annualized profit; it annualizes small nominal spreads over short holding periods and should not be read as a long-horizon return.

Figure 10 adds two descriptive views of the screened bonding trades. The scatter plot relates total profit per market, for the top 100 bonding-trade makers, to the number of bonding-trade makers active in that market. The histogram reports the distribution of bonding-trade APYs: each bin is an APY

range and its height counts how many trades fall in that range. The median bonding-trade APY is 6.22%, while the mean is 29.08%.



(a) Total profit per market and number of bonding-trade makers for the top 100 bonding-trade makers. (b) Bonding-trade APY distribution. Histogram bins are APY ranges; counts are numbers of trades.

Figure 10: Additional descriptive views of settlement-liquidity provision. The left panel plots total profit per market against the number of bonding-trade makers active in each market. The right panel shows the frequency distribution of bonding-trade APYs for the top 100 bonding-traders by transaction-count; the median is 6.22% and the mean is 29.08%.

9.4 Supplementary AAVE–ASW Comparison

Figure 11 visualizes the unsmoothed daily relation between external DeFi yields and Polymarket frontier-implied ASWs across market regimes.

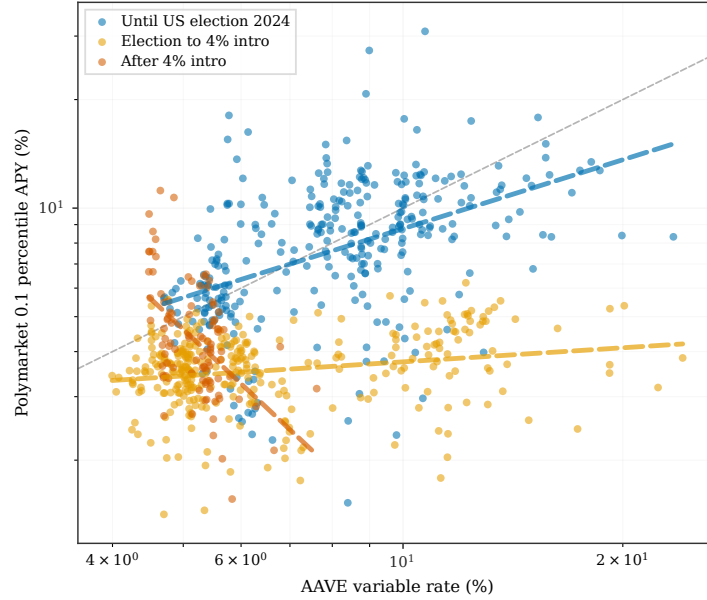


Figure 11: Unsmoothed daily scatter of annualized AAVE variable supply rates and Polymarket daily 0.1-percentile frontier-implied ASWs. Colors distinguish the pre-election period, the post-election/pre-yield-program period, and the period after Polymarket’s 4% yield program. The comparison is descriptive: the two series reflect different liquidity, settlement, and risk structures rather than a direct arbitrage spread.

The scatter reinforces the interpretation from the main text: the relation between external funding benchmarks and frontier-implied ASWs is noisy, regime-dependent, and not reducible to stable arbitrage parity.

9.5 Supplementary Calibration Visualization

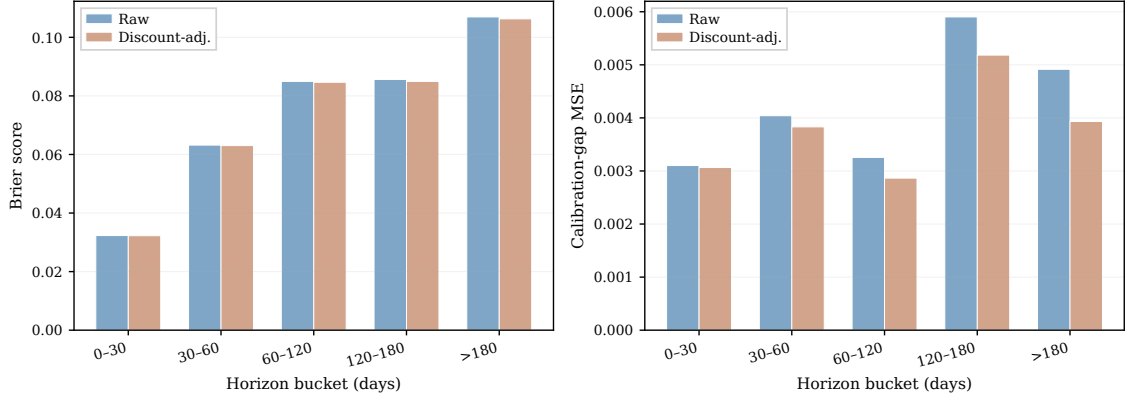


Figure 12: Graphical counterpart to Table 2. Raw Brier scores worsen with settlement horizon. Discount adjustment changes Brier scores only modestly, but more clearly reduces calibration-gap MSE, especially at longer horizons.

Figure 12 clarifies why calibration-gap MSE is the more mechanism-sensitive diagnostic: settlement discounting primarily affects reliability, whereas Brier scores also reflect residual outcome uncertainty.

9.6 Maturity-Gradient Attenuation

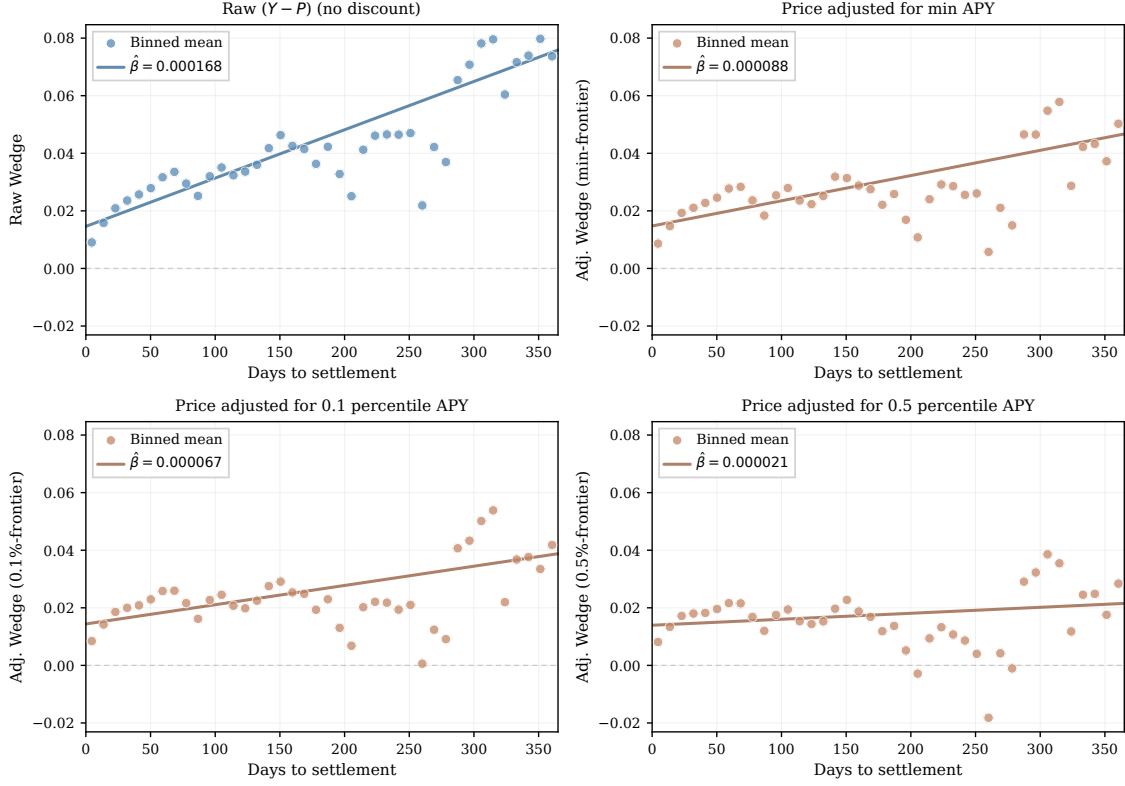


Figure 13: Binned visualization of the regression specifications corresponding to Table 3. The panels plot the raw wedge $X_i - P_{i,t}$ and the adjusted wedges using date-varying discount curves estimated from the minimum-ASW, 0.1-percentile, and 0.5-percentile near-certainty frontiers. Dots are binned means over horizon intervals up to 365 days, while red lines are OLS fits estimated on the full sample. Standard errors are clustered by market.

Figure 13 visualizes the maturity-gradient attenuation underlying Table 3. The raw specification shows that longer-horizon near-certain contracts trade farther below eventual payoff, consistent with settlement-induced discounting increasing with horizon. The discount-adjusted specifications progressively flatten the gradient; attenuation increases monotonically as the lower-tail frontier is estimated more conservatively, with the 0.5-percentile frontier producing an adjusted slope statistically indistinguishable from zero.

9.7 Event-Level Frontier Portability

A potential concern with the attenuation exercise is that the estimated discount curve $\hat{D}(\tau)$ is itself horizon-shaped and estimated from near-certain observations in the same structural environment. A curve estimated from that environment may mechanically attenuate the same horizon pattern it summarizes. We therefore repeat the main wedge regression using event-level cross-fitted discount curves: events are split into five folds, each frontier is estimated on the other four folds, and the resulting curve is applied only to held-out events before pooling the adjusted observations.

Table 7: All-event and held-out-event attenuation of the near-certainty horizon gradient. The raw slope is $\hat{\beta}_{\text{raw}} = 0.00016780$. Adjusted prices are capped at one after frontier adjustment. P-values are clustered by market.

Specification	Frontier	Adjusted slope	Slope reduction	p-value
All events	Minimum	0.00008757	47.8%	0.0079
All events	0.1 percentile	0.00006686	60.2%	0.0425
All events	0.5 percentile	0.00002076	87.6%	0.5285
Held-out events	Minimum	0.000073	56.4%	0.026
Held-out events	0.1 percentile	0.000055	67.2%	0.094
Held-out events	0.5 percentile	0.000012	92.9%	0.717

The held-out-event estimates are the key portability test. Frontiers estimated from other event sets reduce the held-out horizon gradient by 56.4% to 92.9%, slightly more than the corresponding all-event adjustments. Because attenuation does not weaken when the adjusted events are excluded from frontier estimation, the frontier is more consistent with a common platform-level settlement component than a purely same-sample statistical artifact.

9.8 Structural negRisk Payoff Identity

Claim (negRisk payoff identity). Consider one event with n mutually exclusive and exhaustive outcomes and realized outcome $\omega \in \{1, \dots, n\}$. Let

$$Y_k(\omega) = \mathbf{1}\{\omega = k\}, \quad N_k(\omega) = 1 - \mathbf{1}\{\omega = k\}.$$

For any subset $S \subseteq \{1, \dots, n\}$ with cardinality $m = |S|$, the NO-basket satisfies the exact payoff equivalence

$$\sum_{k \in S} N_k \equiv (m - 1) \cdot \mathbf{1} + \sum_{j \notin S} Y_j. \quad (1)$$

Equation (1) decomposes a NO basket into a deterministic component $(m - 1)\mathbf{1}$ and a residual YES basket on the complementary outcomes. This identity underlies negRisk conversion: it allows an economically equivalent position to be transformed into cash-like collateral plus only the remaining complementary outcome exposure, compressing the capital that remains locked until settlement. In the common case $m = n - 1$, the residual exposure is a single YES claim.

Proof. Define $\Pi_S := \sum_{k \in S} N_k$. Its payoff is

$$\Pi_S(\omega) = \sum_{k \in S} (1 - \mathbf{1}\{\omega = k\}) = m - \mathbf{1}\{\omega \in S\} = (m - 1) + \mathbf{1}\{\omega \notin S\}.$$

Since outcomes are mutually exclusive and exhaustive,

$$\mathbf{1}\{\omega \notin S\} = \sum_{j \notin S} Y_j(\omega).$$

Substitution gives Equation (1). $\square$

9.9 Fee-Adjusted negRisk Conversion Bounds

Equation (1) is an exact payoff identity. Conversion fees do not change that identity, but they reduce the net cash-like and residual YES outputs. The result is an idealized conversion-implied lower bound.

Exact fee perturbation. Let $\phi_c \in [0, 1)$ denote the cash-leg fee rate and $\phi_y \in [0, 1)$ the YES-leg fee rate. A conversion of the NO basket $\{N_k\}_{k \in S}$ with $|S| = m$ yields net outputs

$$\sum_{k \in S} N_k \longrightarrow (m - 1)(1 - \phi_c) \cdot \mathbf{1} + (1 - \phi_y) \sum_{j \notin S} Y_j. \quad (2)$$

In bips-based implementations, a single parameter maps to $\phi = \text{feeBips}/\text{FEE_DENOMINATOR}$; when the split is unknown, $\phi_c = \phi_y = \phi$ is the reduced-form case.

Reduced-form price bound. Let $P(N_k, t)$ and $P(Y_j, t)$ denote time- t prices and normalize USDC to one. Abstracting from execution frictions other than conversion fees, buying the NO basket and converting gives

$$\sum_{k \in S} P(N_k, t) \geq (m-1)(1-\phi_c) + (1-\phi_y) \sum_{j \notin S} P(Y_j, t). \quad (3)$$

Under the reduced-form settlement-discount model $P(Y_j, t) = p_j(t)D(\tau)$, near-eliminated subsets with $p_S(t) := \sum_{k \in S} p_k(t) \approx 0$ satisfy

$$\sum_{k \in S} P(N_k, t) \gtrsim (m-1)(1-\phi_c) + (1-\phi_y)D(\tau). \quad (4)$$

Canonical n -market line. In the common case $m = n-1$, let $\bar{P}_N(\tau; n)$ denote the average price of the $n-1$ NO legs in the basket. Equation (4) gives the fee-adjusted conversion-implied line

$$\bar{P}_N(\tau; n) \gtrsim \frac{(n-2)(1-\phi_c) + (1-\phi_y)D(\tau)}{n-1}. \quad (5)$$

Relative to the fee-free line $[(n-2)+D(\tau)]/(n-1)$, fees lower the price floor by $[(n-2)\phi_c + \phi_y D(\tau)]/(n-1)$ and therefore raise the annualized implied yield.

Short-horizon implication. A first-order expansion near par clarifies the maturity dependence of fees. With $D(\tau) \approx 1 - r\tau$, the approximate continuously compounded implied rate is

$$r_{\text{imp}}(\tau) \approx \frac{r}{n-1} + \frac{(n-2)\phi_c + \phi_y}{(n-1)\tau}. \quad (6)$$

The first term is the compressed settlement-discount rate; the second is the fee term. Because the fee term scales as $1/\tau$, conversion fees matter disproportionately at short maturities, where small price wedges annualize sharply.

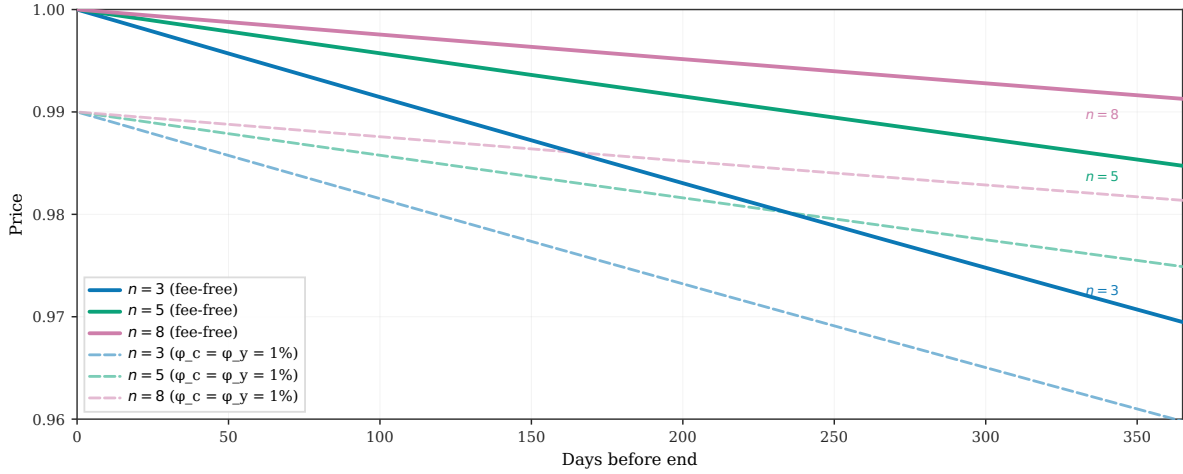


Figure 14: Fee-adjusted theoretical negRisk price floors. Solid colored curves show fee-free conversion-implied bounds for events with n mutually exclusive outcomes; dashed colored curves set $\phi_c = \phi_y = 1\%$. Larger n and lower fees move the floor closer to par.

Figure 14 isolates the fee channel in the negRisk mechanism. A NO basket converts into a cash-like component, subject to successful conversion and settlement, plus a residual YES claim. As n rises, the residual exposure becomes smaller relative to the cash-like part and the bound moves closer to par. Fees relax this idealized lower bound: the cash-leg fee loads on the deterministic component, while the YES-leg fee loads only on the discounted residual claim. In Polymarket's deployed NegRiskAdapter, a single market-specific `feeBips` parameter is applied to both outputs of conversion, making the symmetric case $\phi_c = \phi_y = \phi$ the closest reduced-form representation.⁵

⁵Polymarket NegRiskAdapter repository: <https://github.com/Polymarket/neg-risk-ctf-adapter>.

9.10 Taker-Fee-Adjusted Execution Bounds

The previous subsection studies fees charged inside the negRisk adapter. A separate friction arises before conversion: if the trader acquires the required legs with marketable orders, Polymarket taker fees raise the effective acquisition cost. This lowers the nominal posted-price bound at which the conversion trade breaks even.

For politics markets, let the taker-fee rate be $f = 0.04$ [Pol26c]. A marketable trade at price p pays per-share fee $fp(1 - p)$, so the effective acquisition price is

$$p^{\text{eff}} = p + fp(1 - p).$$

In a near-tail conversion, the same fee formula can be read as applying to a low-priced YES leg near $p \approx 0.01$, or equivalently to the complementary NO leg near $1 - p \approx 0.99$. The traded-leg interpretation changes, but the fee schedule is symmetric in p and $1 - p$. The per-share taker fee is therefore approximately

$$fp(1 - p) \approx 0.04 \cdot 0.01 \cdot 0.99 \approx 0.000396.$$

For a basket S with $|S| = m$, the taker-executable conversion bound is

$$\sum_{k \in S} [P(N_k, t) + fP(N_k, t)(1 - P(N_k, t))] \geq (m - 1)(1 - \phi_c) + (1 - \phi_y) \sum_{j \notin S} P(Y_j, t).$$

Rearranging gives the nominal posted-price condition

$$\sum_{k \in S} P(N_k, t) \geq (m - 1)(1 - \phi_c) + (1 - \phi_y) \sum_{j \notin S} P(Y_j, t) - \sum_{k \in S} fP(N_k, t)(1 - P(N_k, t)).$$

In the canonical case $m = n - 1$, using the near-tail approximation $P(N_k, t) \approx 0.99$, the average nominal NO-price bound becomes

$$\bar{P}_N^{\text{taker}}(\tau; n) \gtrsim \bar{P}_N^{(\phi)}(\tau; n) - f \cdot 0.99 \cdot 0.01,$$

where

$$\bar{P}_N^{(\phi)}(\tau; n) = \frac{(n - 2)(1 - \phi_c) + (1 - \phi_y)D(\tau)}{n - 1}.$$

For politics markets,

$$\bar{P}_N^{\text{taker}}(\tau; n) \gtrsim \bar{P}_N^{(\phi)}(\tau; n) - 0.000396.$$

Taker fees therefore perturb the conversion-implied floor by about four basis points in price space near par. The level effect is small, but it is exactly the type of wedge that can become salient in short-horizon APYs because annualization scales a fixed price cost over a shrinking settlement horizon.

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